Network Science

Lecture 08: Network Dynamics

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Speaker notes

This lecture makes the final conceptual shift of the course: from structure to dynamics. The graph is no longer just an object to analyze; it becomes the medium through which processes move. The didactic rhythm is deliberately repeated: intuitive example first, then theory, then a practical example, then applications. Students should leave with a modeling vocabulary: if something spreads, first ask what the node states are, what makes a node change state, whether the state sticks, and which network structures help or block the process.

From Structure to Dynamics

Lectures 03-07 analyzed networks as static objects: ties, bridges, communities, balance, short paths, hubs, and small-world shortcuts.

Today we ask a different question:

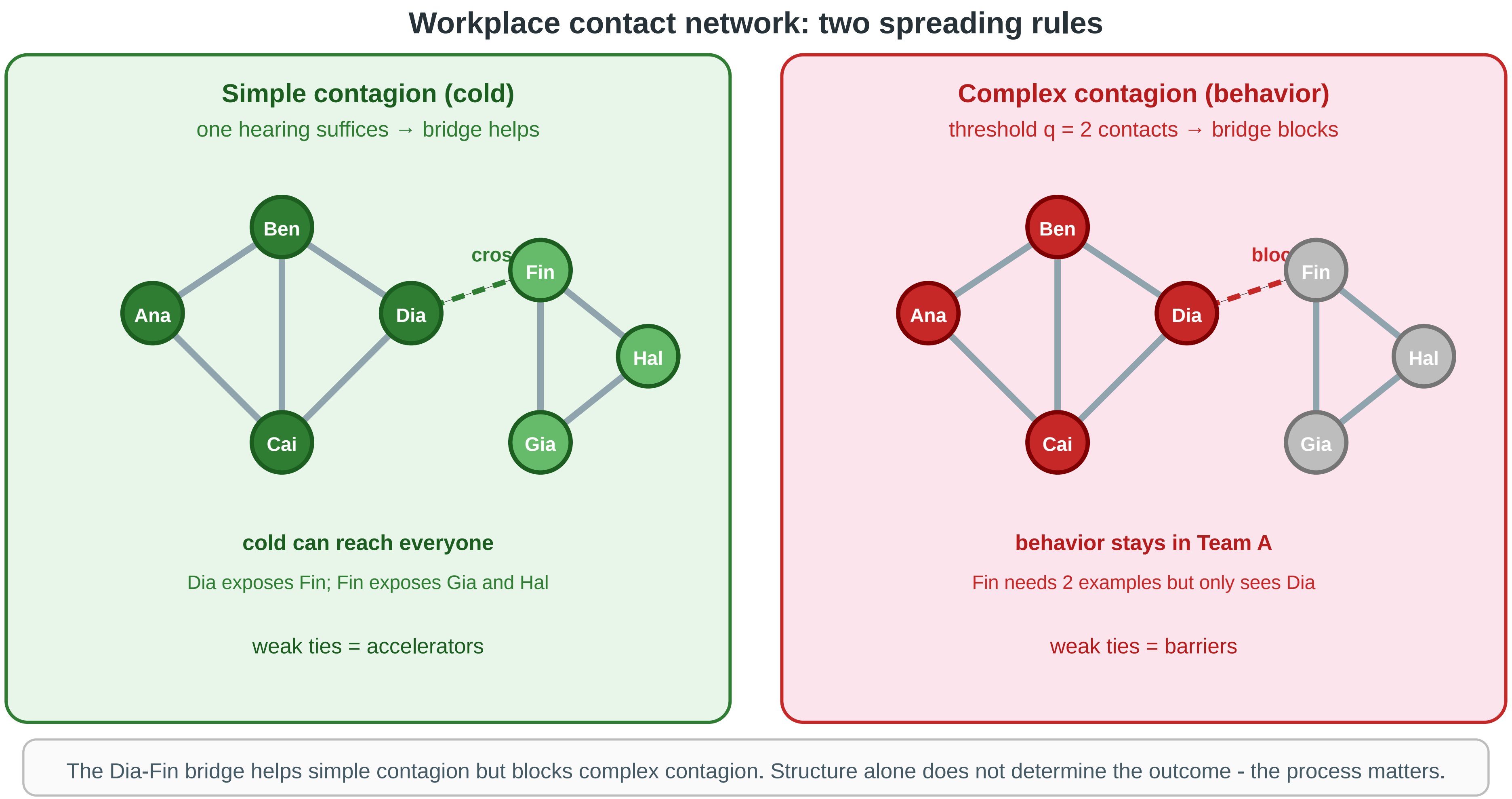
> **What happens *on* the network?**

A cold, a rumor, PageRank score, a product failure, and a health behavior can all spread over edges. But they do not follow the same rule.

Speaker notes

Keep this opening simple. Students have already seen the structural vocabulary. The new idea is that structure is only half the model. The process rule decides whether a bridge is useful, whether a hub is dangerous, whether clustering helps, and whether time ordering matters.

Intuition: Same Network, Different Processes



Cold: one infectious contact can be enough, so the Dia-Fin bridge can carry the cold to the second team.

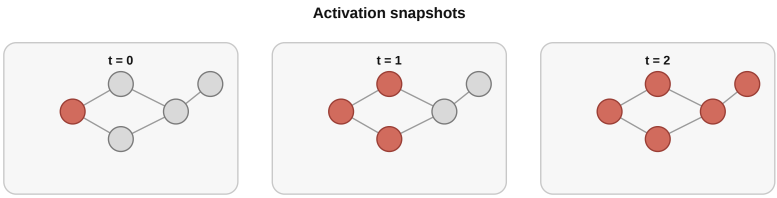
Health behavior: Fin may need to see two peers join before changing behavior, so the same bridge is too thin. The behavior stalls at the community boundary.

Speaker notes

This figure remains the anchor for the lecture. The key lesson is not just "things spread"; it is "the same edge can have opposite effects for different processes." In the language of the lecture, the cold is simple contagion, while the health behavior is threshold-based complex contagion.

Four Questions

1. **What spreads?** Signal, probability mass, disease, adoption, failure?
2. **What changes a node?** One exposure, repeated exposure, a threshold, or a flow equation?
3. **Does the state stick?** Forever, temporarily, or not at all?
4. **Which topology helps?** Hubs, bridges, clusters, shortcuts, or communities?



Red nodes are active; each step spreads activity to reachable neighbors.

Speaker notes

These four questions replace the old quick-check structure. They also provide the repeated modeling checklist for all applications. For every new spreading problem, students should identify the object being spread, the state-transition rule, the state memory, and the relevant topology.

The Process-Structure Gap

The course has accumulated several gaps between clean theory and real data. L08 adds the **process-structure gap**.

Lecture	Gap type	Core issue
L03-L04	Computational	exact ideals vs. feasible proxies
L05	Causal	snapshots do not identify mechanisms
L06	Structural	complete-graph theory vs. sparse data
L07	Navigational	short paths exist but may not be findable
L08	**Process-structure**	same graph, different outcomes for different processes

Speaker notes

This slide connects the lecture to the course arc. Avoid spending long on the taxonomy. The important idea is that there is no universal "best" structural feature for spreading. A bridge helps a cold but can block a behavior that needs reinforcement. A hub accelerates simple contagion but may be less useful for threshold behavior if its neighbors are isolated from each other.

The Process-Structure Gap

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L08	**Process-structure**	same graph, different outcomes for different processes

Network structure is necessary but not sufficient. A spreading model needs both a graph and a rule.

Takeaway

> Network dynamics ask how states, signals, behaviors, or risks move over edges. The right model depends on the spreading rule: flow, cascade, threshold, infection, recovery, persistence, and timing.

Basic Network Diffusion Models

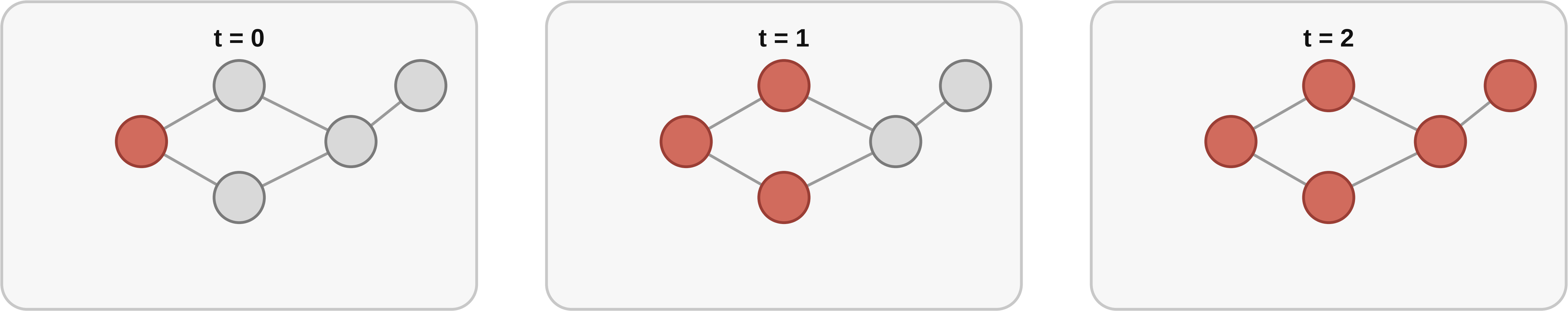
Guiding Question: How can we model the spread of a signal, activation, or influence before we specialize to epidemics?

Speaker notes

This module comes before SIR because students need a broader concept of diffusion first. Epidemic models are then a special case with disease-like states and recovery. The sequence is example first, then theory, then practical comparison, then applications.

Intuitive Example: A Message Starts at A

Activation snapshots



Red nodes are active; each step spreads activity to reachable neighbors.

> A message starts at one seed node. It may move like a wave, leak like probability mass, trigger each neighbor independently, or require social reinforcement.

The graph gives possible channels. The diffusion rule says which channels are actually used.

Speaker notes

Use a simple verbal trace: A tells B and C; B and C may tell D; D may tell E. Then ask: is D active because one message is enough, because two messages reinforced each other, or because some fraction of attention flowed there? Those are different models.

Case 1: Deterministic Reachability

Start with the simplest possible rule:

> If a node is active, it activates **all** of its inactive neighbors in the next step.

This is the BFS idea from L02 turned into a spreading process.

Time	Active nodes
$t = 0$	A
$t = 1$	A, B, C
$t = 2$	A, B, C, D
$t = 3$	A, B, C, D, E

Speaker notes

This is the standard classroom baseline because it separates topology from uncertainty. If a path exists, the signal eventually follows it. It is useful for reachability, shortest spreading time, and dependency analysis, but it is too strong for many human processes.

Why Reachability Is Too Strong

In real spreading, two things often fail:

1. **Not every neighbor is contacted.** Ana may tell only one person, not everyone.
2. **Not every contacted neighbor responds.** A person may ignore a message or not click a link.

> So we need a softer model: instead of activating all neighbors, let the spread choose paths probabilistically.

Application. Browsing, attention, and search behavior: users follow one link at a time, not every outgoing link at once.

Speaker notes

This slide motivates random walks. It should feel obvious: deterministic spread is useful but unrealistic when attention is limited. A student should understand why the next model exists before seeing the matrix notation.

Case 2: Random-Walk Diffusion

A random walker sits on one node. At each step, it chooses one outgoing edge and moves along it.

If node j has out-degree d_j, then each outgoing neighbor receives probability $1/d_j$.

For the transition matrix P:

- $P_{ij} = 1/d_j$ if there is an edge $j \to i$
- $P_{ij} = 0$ otherwise

P is a **transition matrix**: column j says where probability moves next from node j.

For $A \to B, A \to C, B \to D, C \to D$:

P	from A	from B	from C	from D
to A	0	0	0	0
to B	1/2	0	0	0
to C	1/2	0	0	0
to D	0	1	1	0

From A, half the mass goes to B and half to C.

Speaker notes

This slide defines transition matrix mathematically and visually. The convention here is column-stochastic: $x_{t+1} = Px_t$, so column j distributes the mass currently at node j. Mention that some textbooks use the transpose convention; the modeling idea is the same.

Random-Walk Theory: Reading the State Vector

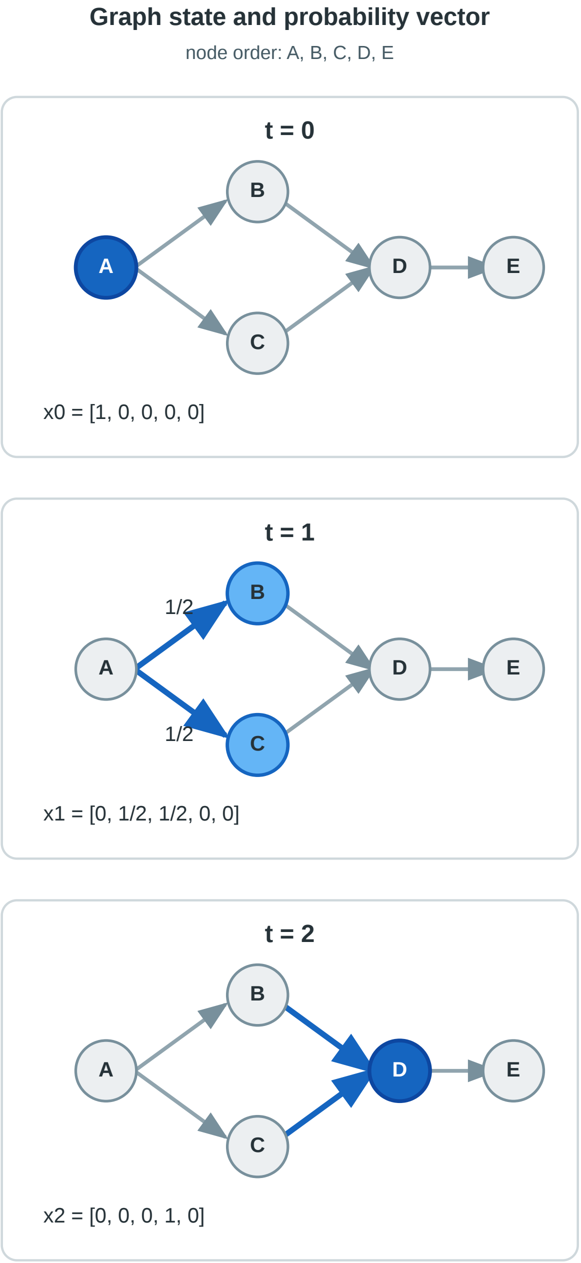
The probability state vector x_t has one entry per node.

If the node order is (A, B, C, D, E), then

$$x_t = [0, 1/2, 1/2, 0, 0]$$

means: at time t, the walker is at B with probability $1/2$ and at C with probability $1/2$.

The entries sum to 1 when probability is conserved.



Speaker notes

Use this slide to make the vector concrete. The vector is not a list of active nodes. It is a probability distribution over possible locations of the walker. The entries sum to 1 when probability is conserved. In this example the mass starts at A, splits evenly to B and C, and then recombines at D.

Random-Walk Theory and Application

If x_t stores the probability mass at time t, then:

$$x_{t+1} = P x_t.$$

> The process no longer asks "which nodes become active?" It asks "where is the probability mass after repeated movement?"

Example. Starting from A, probability splits between B and C, then recombines at D.

Application. Web browsing, PageRank-like importance, traffic flow, spreading activation in search and recommendation.

Speaker notes

This distinguishes binary activation from graded diffusion. Students should see that random-walk diffusion is not an epidemic model. It is a flow model.

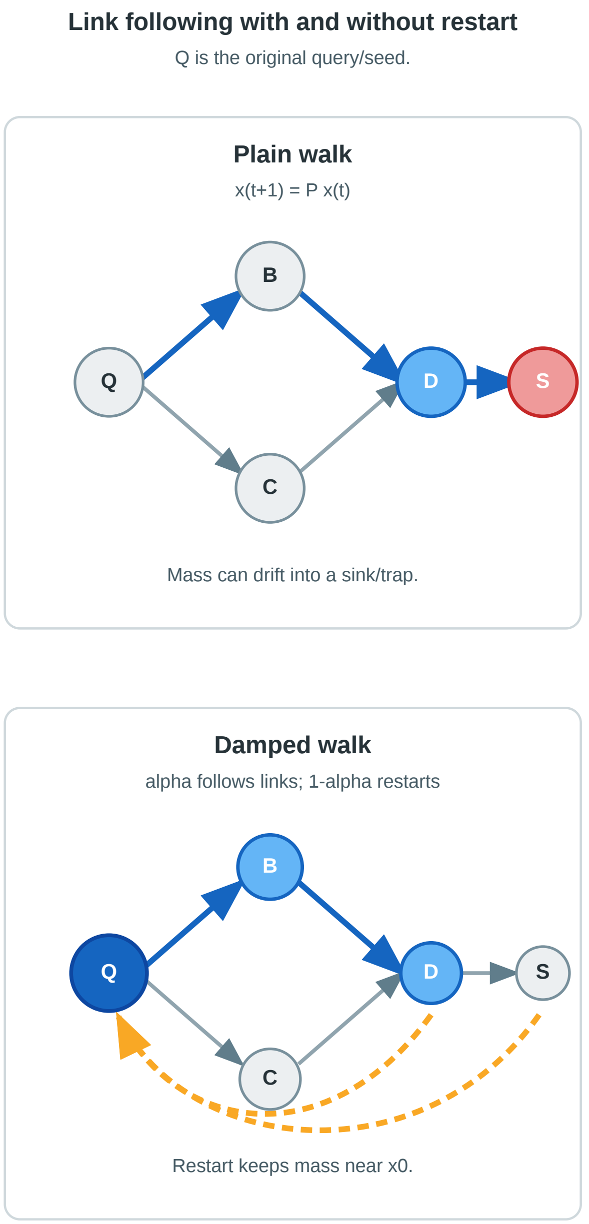
Why Plain Random Walks Need Damping

A plain random walk can drift too far from the original topic or get trapped in a sink.

> Imagine searching for "network diffusion." Following links forever may eventually land on a barely related page. A useful model should sometimes return to the original query or seed node.

This return is called **damping**, **restart**, or **teleportation**, depending on the application.

Application. Personalized search: explore nearby pages, but keep pulling attention back to the user's original interest.



Speaker notes

This is the intuitive explanation before the formula. Damping solves two problems: attention should decay with distance, and directed graphs can have sinks or traps. PageRank used the same idea globally; personalized PageRank uses a query- or seed-specific restart vector.

Case 3: Damped Diffusion

A damped diffusion mixes two actions:

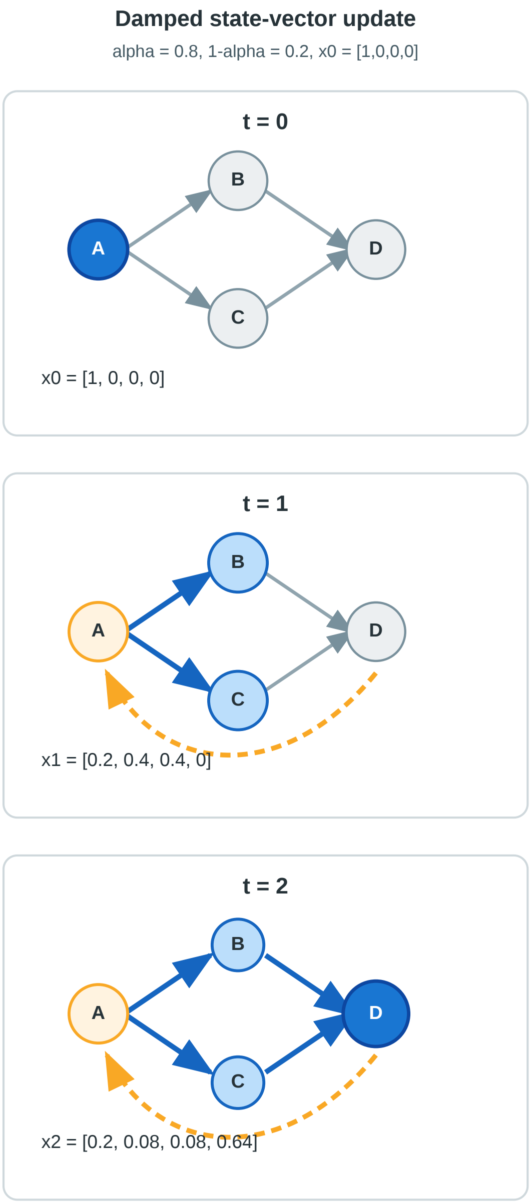
1. **follow the network** with probability α
2. **restart** at the seed distribution x_0 with probability $1-\alpha$

$$x_{t+1} = \alpha P x_t + (1-\alpha)x_0, \qquad 0 < \alpha < 1.$$

x_0 is the **restart distribution**: it says where the walk returns when it restarts. For personalized search, x_0 is concentrated on the query or seed nodes.

α is the **link-following probability**. Large α: explore farther. Small α: stay close to x_0.

PageRank uses this random-surfer idea for ranking Web pages (Brin & Page, 1998).



Speaker notes

This should now feel motivated rather than appearing as a trick. The restart vector x_0 can be uniform, as in classic PageRank, or concentrated on a seed/query for personalized PageRank.

Why Random Walks Are Still Too Smooth

Random walks describe attention or probability mass. But many social processes are **events**:

- a person shares a post or does not
- a customer buys or does not
- a device fails or does not

> For event-like spread, we need a model where each exposure can succeed or fail.

That gives the **independent cascade** model.

Speaker notes

This is the transition from continuous/graded diffusion to binary event diffusion. It also explains why independent cascade is not just a random walk with different notation. A random walk moves one unit of probability mass along one outgoing edge per walker step. Independent cascade models event attempts along edges.

Case 4: Independent Cascade

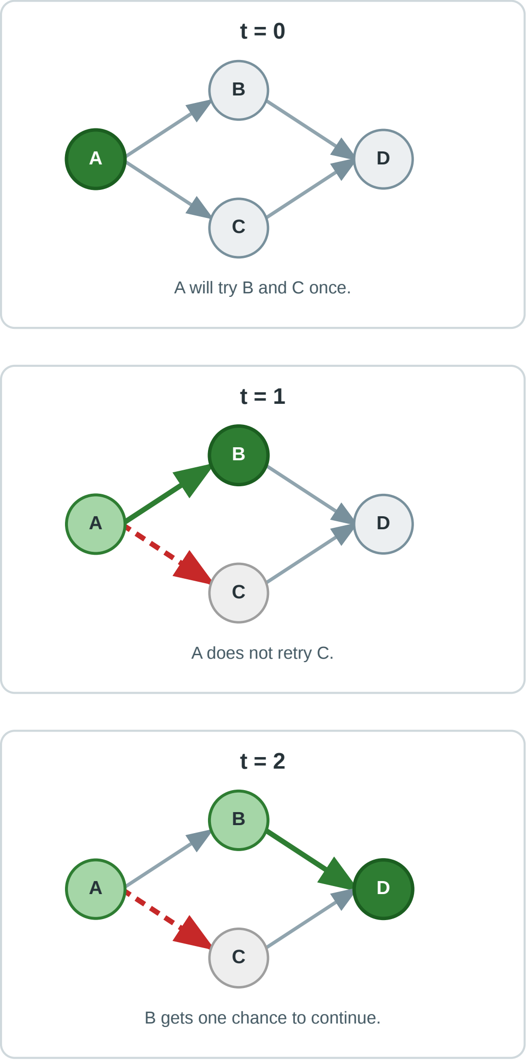
Each newly active node gets **one chance** to activate each inactive neighbor.
For an edge $u \to v$, activation succeeds with probability p_{uv}.

Independent cascade is **not** a random walk. The active node does **not** choose one outgoing edge. It tries **all** outgoing edges once, and each edge succeeds or fails independently. Therefore one active node can activate many neighbors in the same step, or none.

Example. Ana posts a link. Ben clicks and shares with probability 0.3; Cai clicks and shares with probability 0.1. If Ben does not share now, Ana does not keep retrying forever.
Application. Viral marketing, information cascades, link sharing, one-shot recommendations. Kempe, Kleinberg & Tardos (2003) use this model for influence maximization.

One chance per outgoing edge

each edge succeeds or fails independently



Speaker notes

The "one chance per edge" assumption is the key simplification. If Ana has edges to Ben, Cai, and Dia, Ana attempts all three edges once. These attempts are independent Bernoulli trials: Ben may activate, Cai may not, Dia may activate. There is no conservation of probability mass and no rule that all unchosen edges go to zero because no single edge is chosen. It makes sense for a one-time exposure but not for repeated pressure or coordination. That limitation motivates the next model.

Why One Exposure Is Sometimes Not Enough

Some behaviors are not triggered by one contact:

- joining a protest may require seeing several friends join
- registering for a health program may require social proof
- changing a habit may require repeated reinforcement

> Here the important signal is not "did one edge transmit?" but "how much of my neighborhood is already active?"

That gives the **linear threshold** model.

Speaker notes

Use the health-program example to connect smoothly to Centola later. This avoids the unclear workplace-tool framing while preserving the important complex-contagion concept.

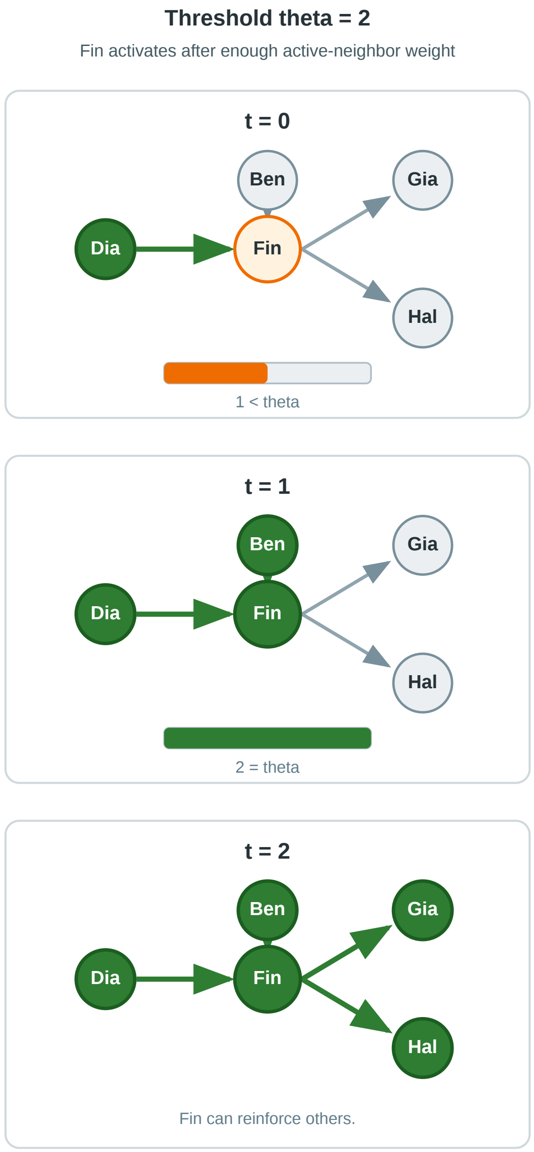
Case 5: Linear Threshold

Node v activates when enough weighted active neighbors point to it:

$$\sum_{u \in N(v)} w_{uv} x_u \geq \theta_v.$$

$x_u = 1$ means neighbor u is active; θ_v is v's adoption threshold.

Example. Fin joins a health program only after seeing at least two close colleagues join.
Application. Behavior change, norm adoption, political mobilization, technologies that need coordination.
Granovetter (1978) formalized threshold models of collective behavior; Kempe, Kleinberg & Tardos (2003) use linear threshold models for influence spread.



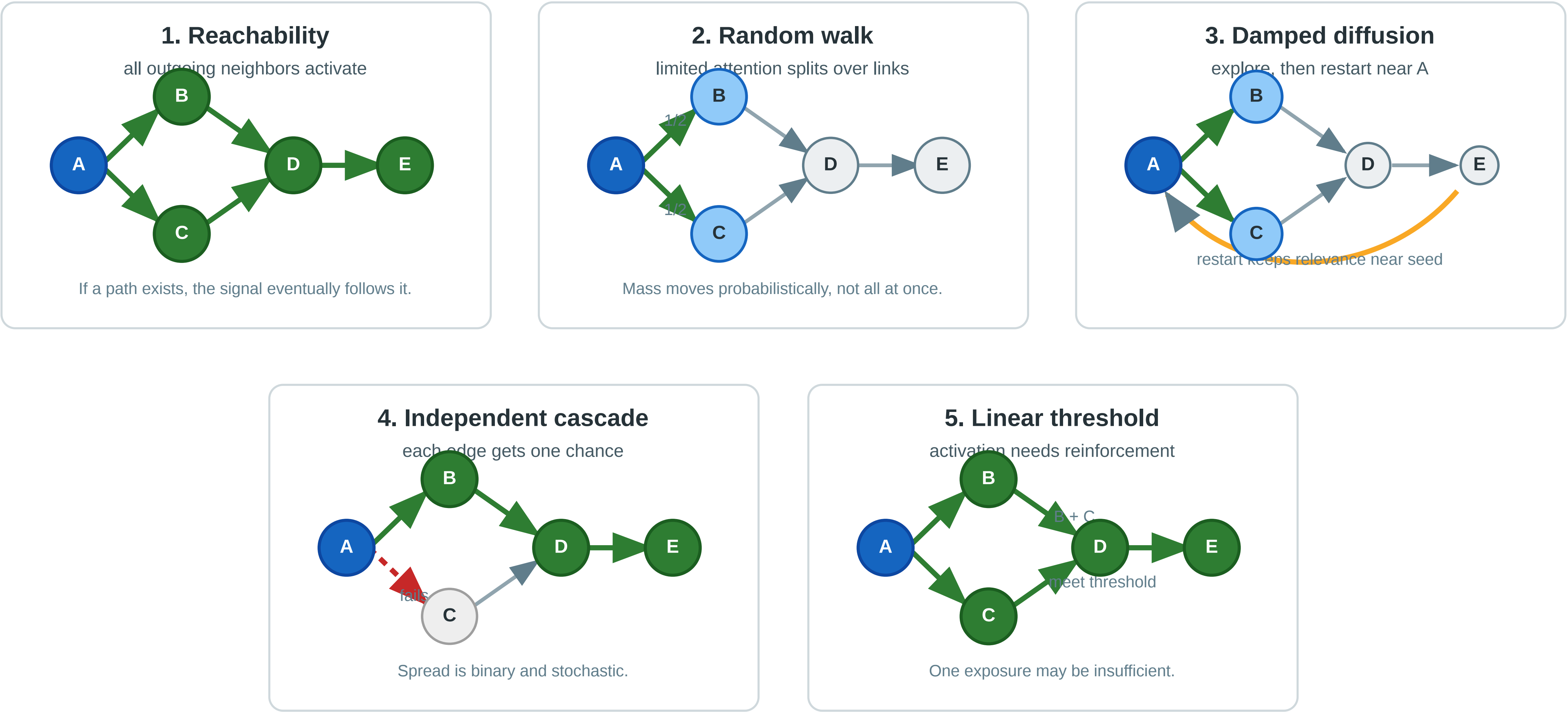
Speaker notes

The weights can represent tie strength or exposure intensity. If all weights are equal, the rule becomes a fraction or count of active neighbors. This is the bridge to complex contagion: reinforcement from multiple neighbors is the mechanism.

Practical Comparison: Same Graph, Different Rules

Same graph, five different diffusion rules

A starts the process. The graph is fixed; only the spreading rule changes.



The graph is fixed. Reachability activates every reachable node; random walks move probability mass; damping keeps diffusion close to the seed; independent cascade makes each edge uncertain; threshold diffusion requires reinforcement.

Speaker notes

This figure is the visual synthesis for Module 1. Walk from left to right: first the unrealistic but clear baseline, then limited attention, then restart/damping, then stochastic one-shot transmission, then reinforcement. The students should see that the same graph does not imply the same spreading outcome.

Applications: Which Diffusion Rule Fits?

Application	Natural model	Why
Web ranking	damped random-walk diffusion	attention flows through links with teleportation
Viral post	independent cascade	one exposure can trigger sharing
Health behavior	threshold model	peers provide social proof
Search over related concepts	diffusion with decay	nearby concepts should count more
Failure propagation	cascade model	one failed dependency can trigger another

> The modeling choice is substantive. Choosing a rule means making a claim about how exposure changes behavior.

Speaker notes

This slide should slow students down. Diffusion models are not just formulas; they encode assumptions about human behavior, technical dependencies, or information flow. A health-behavior intervention should usually not be modeled as PageRank, and PageRank should not be interpreted as an epidemic.

Module Takeaway

Basic diffusion models differ in what a node state means: binary activation, probability mass, graded attention, one-shot influence, or threshold behavior. Epidemic models are one important special case, not the whole category.

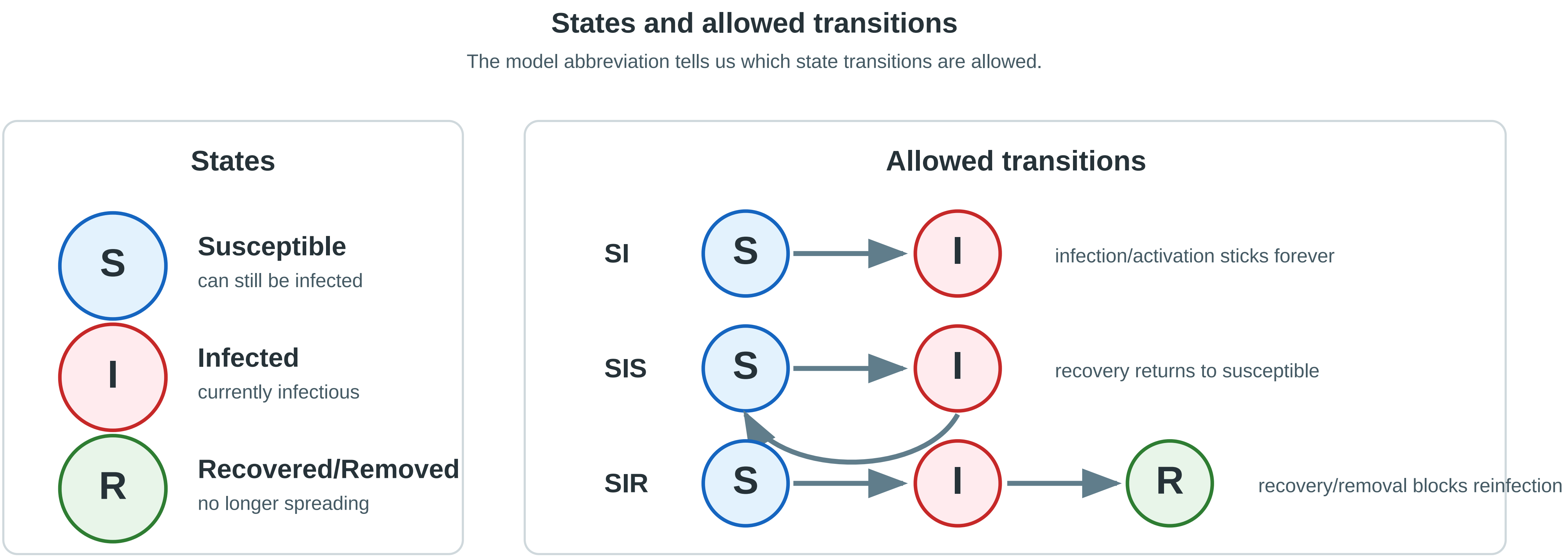
Epidemic Models and State Memory

> **Guiding Question:** What changes when diffusion is modeled as infection, recovery, immunity, or reinfection?

Speaker notes

State memory asks what happens after exposure: does the state remain, disappear, or become resistant to future infection? That single modeling choice separates SI, SIS, and SIR.

Epidemic State Vocabulary



Bridge from basic diffusion: inactive -> active is SI written in epidemic notation.

> The acronym names the allowed path through states: **SI** means $S \to I$, **SIS** means $S \to I \to S$, and **SIR** means $S \to I \to R$.

Speaker notes

Use this as the vocabulary slide. S means susceptible: can still be infected. I means infected and infectious: currently able to transmit. R means recovered or removed: no longer infectious and no longer susceptible in the model.

Case 1: SI

> **SI:** $S \to I$
>
> Susceptible nodes become infected. There is **no recovery** and no removal.

Intuition. Infection or activation accumulates. Once a node enters I, it stays in I and can keep transmitting.

Example. On a path, an infection starting at B grows outward until all reachable nodes are infected.

Application. Cumulative awareness, irreversible adoption, permanent compromise, reachability with epidemic labels.

Speaker notes

The key teaching point: SI is the basic model. It is useful when the state is sticky and no recovery process matters for the question.

SI: Formal Network Update

Let $G = (V, E)$ and let $X_i(t) \in S, I$ be the state of node i at time t.

For a susceptible node i, define its infected neighbors:

$$N_I(i, t) = j \in N(i) : X_j(t) = I.$$

With edge transmission probability β, the discrete-time update is:

$$\Pr[X_i(t+1) = I \mid X_i(t) = S] = 1 - (1 - \beta)^{|N_I(i,t)|}.$$

> The equation says: node i becomes infected unless all currently infected neighbors fail to transmit.

> Algorithmically, this is a **stochastic simulation**: at each step, every infected node tries to infect susceptible neighbors. Infected nodes never leave I.

> This is the epidemic version of **Case 1: Deterministic Reachability** if activation is certain; with probabilistic one-shot edge attempts, it becomes close to **Case 4: Independent Cascade**.

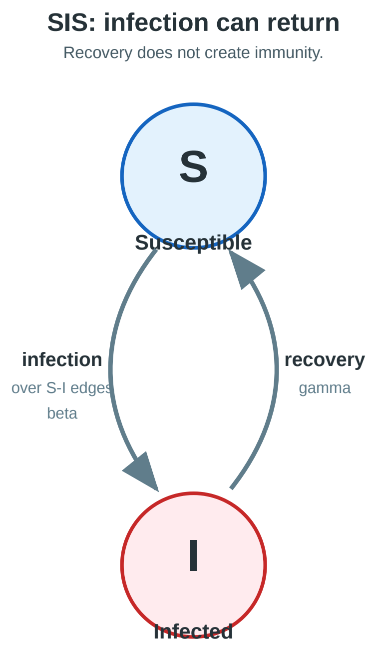
Speaker notes

This is the first place where students see how the state-machine view becomes code. The probability is easier to explain as the complement: no infected neighbor succeeds with probability $(1 - \beta)$ per infected neighbor. If there are many infected neighbors, the infection probability increases.

Case 2: SIS

> **SIS:** $S \to I \to S$
> Infected nodes recover, but recovery returns them to the susceptible state.

Intuition. The infection can come back. A node that recovered is not protected.
Example. In a small group, Ana recovers from a cold but later catches it again from Ben.
Application. Recurring infections without lasting immunity, malware that can be removed and reinstalled, repeated misinformation exposure.



Speaker notes

SIS changes the question from "how far did it ever spread?" to "can it persist?" Reinfection is the defining mechanism. Hubs can be especially important because they can keep reintroducing infection to many neighbors.

SIS: Formal Network Update

SIS uses the same infection step as SI, but infected nodes can recover:

$$\Pr[X_i(t+1) = I \mid X_i(t) = S] = 1 - (1-\beta)^{|N_I(i,t)|}$$

$$\Pr[X_i(t+1) = S \mid X_i(t) = I] = \gamma.$$

> γ is the recovery probability per time step: an infected node returns to S with probability γ and remains infected with probability $1-\gamma$.

> Algorithmically: infection and recovery are repeated forever. A simulation stops only when the infection dies out, reaches a time limit, or approaches a stable endemic level.

> **Endemic** means the infection does not disappear, but remains present at a roughly stable nonzero level.

Speaker notes

This slide gives SIS the same level of formality as SIR. Emphasize that SIS is not mainly about final reach, because final reach can be all nodes over a long time even when the current infected fraction is small. The key outputs are persistence, extinction probability, and long-run infected fraction.

Case 3: SIR

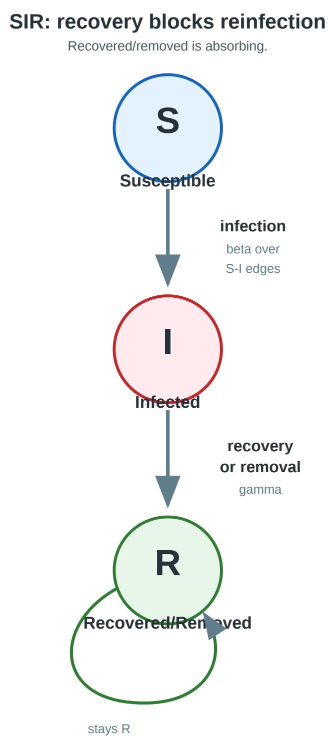
SIR: $S \to I \to R$

Infected nodes recover into a recovered/removed state. They no longer transmit and cannot be infected again in the model.

Intuition. The infection burns through reachable parts of the graph and leaves immune or removed nodes behind.

Example. A one-time infection moves along a path as a wave: current cases are I, past cases are R.

Application. Diseases with immunity, one-shot outbreaks, finite attention cascades, final outbreak size.



Speaker notes

SIR is the model where timing becomes especially visible: infections rise, peak, and fall because the susceptible pool is depleted and recovered nodes block reinfection.

SIR: Formal Network Update

SIR uses infection plus one-way removal:

$$\Pr[X_i(t+1) = I \mid X_i(t) = S] = 1 - (1-\beta)^{|N_I(i,t)|}$$

$$\Pr[X_i(t+1) = R \mid X_i(t) = I] = \gamma.$$

Recovered or removed nodes stay removed:

$$\Pr[X_i(t+1) = R \mid X_i(t) = R] = 1.$$

Speaker notes

SIR has the same local infection rule as SI and SIS. The only difference is the memory state after infection. This is why it is useful to teach the models as variants of one state-transition template rather than as unrelated acronyms.

SIR: What We Compute

SIR is used for time-dependent outbreak quantities:

Quantity	Meaning
$I(t)$	current number or fraction infected
$\max_t I(t)$	peak load
time of peak	when capacity pressure is highest
$R(\infty)$	final outbreak size
extinction probability	probability the outbreak dies early

> On networks, exact closed-form solutions are rare. We usually simulate the stochastic process, approximate it with aggregate equations, or use percolation-style results for final size (Kermack & McKendrick, 1927; Newman, 2002; Van Mieghem, 2014).

Speaker notes

This is the practical answer to "how do you use it?" In computer science terms, SIR is often implemented as an event simulation or synchronous update. Analytical approximations help identify thresholds, peaks, and final sizes, but the exact stochastic network process has a huge state space.

Percolation View of SIR Final Size

Percolation removes the time ordering and asks a simpler reachability question:

> Keep each edge with probability T, where T is the chance that infection would successfully cross that edge before recovery.

How to use it:

- Sample each edge independently: open with probability T, closed with probability $1 - T$.
- In this sampled graph, compute $R(\infty)$ as the nodes reachable from the seed through open edges.
- Repeat both steps many times to estimate expected final size and variability.

$R(\infty)$ is not automatically all nodes; it is the reachable open-edge component from the seed in one sampled graph.

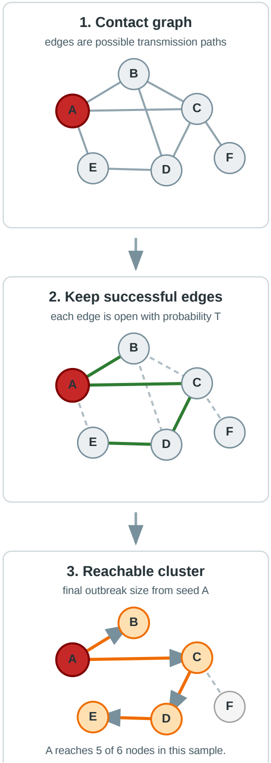
Common ways to compute T:

- **continuous-time rates:** infection and recovery are competing clocks:

$$T = \frac{\beta}{\beta + \gamma}.$$

- **discrete fixed duration:** time is counted in steps, for example days:

$$T = 1 - (1 - p)^{\tau}.$$



Speaker notes

This slide answers whether percolation keeps nodes in states and how T is obtained. T is not a graph property; it comes from the disease/update model. In continuous time, infection across one edge and recovery are competing exponential clocks, so the infection clock wins with probability beta/(beta+gamma). This is not "after 10 days"; it assumes random recovery with rate gamma. If the infectious period is fixed at 10 days, then tau=10 in a discrete daily model with per-day probability p, or a different continuous fixed-duration formula can be used. Other update conventions can give slightly different formulas, so define T consistently with the simulation model. The percolation construction itself does not simulate S, I, and R over time. It samples edges as open or closed transmission opportunities. Once edges are sampled, the final outbreak from a seed is the connected component reachable from that seed through open edges. Repeating the sampling gives an estimate of the distribution of R(infinity). This is why percolation is useful for SIR final size, but not for SIS persistence: SIS has reinfection and no final removed cluster in the same sense.

What Percolation Measures

Percolation is useful when we care about **final reach**, not the full time curve.

Object	Meaning in SIR
open edge	infection would cross this contact before recovery
closed edge	infection would fail to cross this contact
reachable component	nodes eventually infected from the seed
giant component	outbreak can reach a macroscopic part of the network
threshold T_c	graph-level critical transmissibility where large outbreaks become possible

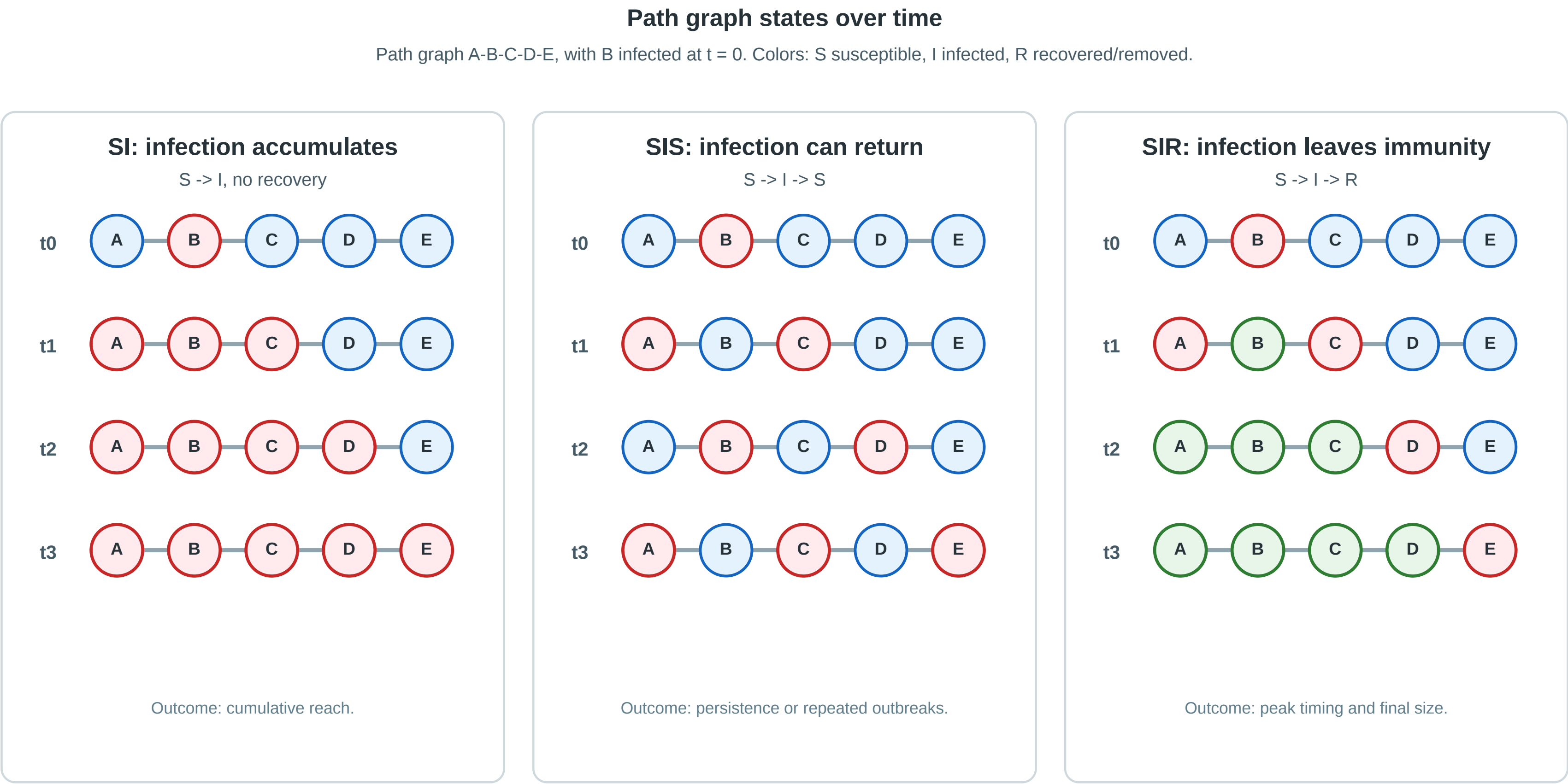
Seed choice depends on the question: use an observed first case, sample random introductions, or compare high-risk seeds such as hubs and bridge nodes.

> Percolation is not a node-state simulation. It is a structural shortcut for estimating final outbreak size under an SIR-like one-shot transmission process.

Speaker notes

This second slide makes measurement explicit. Percolation measures component sizes and reachability under randomly retained edges. T_c is about the graph and transmission model, not about one node. Node-specific questions are different: "if this node is the seed, how large is the reachable open-edge component?" Percolation does not tell us the epidemic peak, time of peak, or duration. Those require time dynamics or simulation.

Practical Example: Same Path, Three Epidemics



The graph and seed are the same. The only difference is the memory rule after infection: stay infected, return to susceptible, or move to recovered/removed.

Speaker notes

Walk through the three panels. SI accumulates infection. SIS allows recurrence and persistence because recovered nodes become susceptible again. SIR creates a moving infection wave and leaves recovered nodes behind.

Aggregate View: From Nodes to Counts

The network model stores a state for every node:

$$X(t) = (X_1(t), X_2(t), \ldots, X_n(t)).$$

The network simulation tracks every node state. The **aggregate view** compresses those node states into counts or fractions:

$$S(t) = |i : X_i(t) = S|, \quad I(t) = |i : X_i(t) = I|, \quad R(t) = |i : X_i(t) = R|.$$

> Aggregation answers "how many?" It no longer answers "which node?", "which bridge?", or "which community?"

Speaker notes

This slide defines aggregate view before showing the SIR curve. The key tradeoff is compression. Aggregates are useful for population-level outcomes, but they deliberately discard the structural path of the outbreak.

Aggregate View: SIS Approximation

A **well-mixed** approximation ignores the graph and assumes every infected node can contact every susceptible node with equal chance.

For SIS, write $I(t)$ for the infected fraction and $S(t) = 1 - I(t)$.

$$\frac{dI}{dt} = \beta SI - \gamma I = \beta I(1 - I) - \gamma I.$$

> This is an **ordinary differential equation (ODE)**: it describes how one quantity changes continuously over time. It is not the closed-form solution; the right-hand side gives the instantaneous rate of change of the infected fraction.

> Infection grows through βSI and shrinks through recovery γI. In the simplest well-mixed form, persistence requires roughly $\beta/\gamma > 1$.

Speaker notes

This slide explains "aggregate approximation" after the aggregate-count slide. It compresses the node-level stochastic process into one population-level variable, I(t). Well-mixed means the model no longer knows the network edges. Because SIS has no removed state, S(t) can be replaced by 1-I(t). The formula shown is the ordinary differential equation, not an explicit solution for I(t).

Aggregate View: The SIR Curve

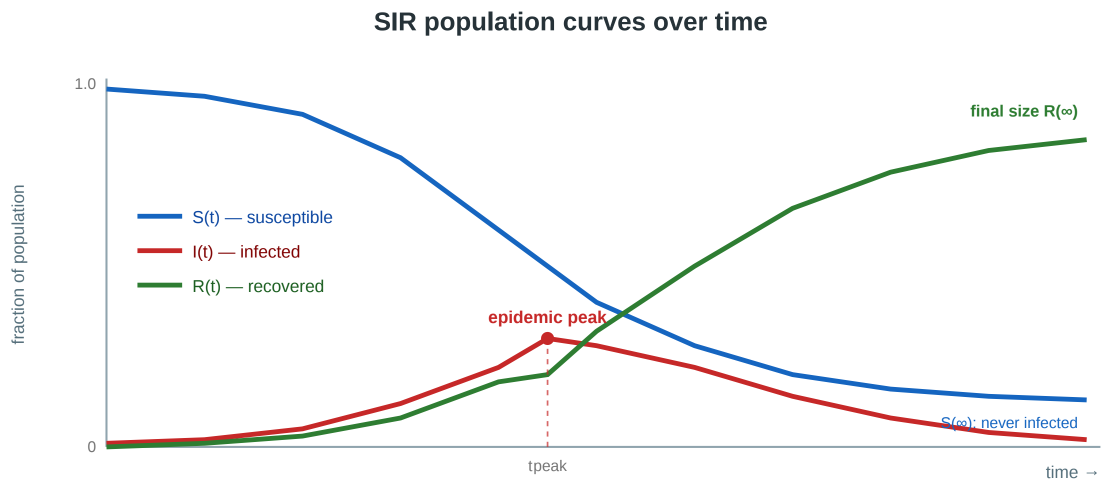
In the well-mixed SIR approximation:

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I.$$

β is the infection rate, γ is the recovery rate; assumes homogeneous mixing, not explicit edges.



Read the chart as three aggregate time series: $S(t)$ falls as susceptible people are infected, $I(t)$ rises while new infections exceed recoveries and then falls after the peak, and $R(t)$ accumulates everyone who has recovered or been removed. $R(\infty)$ is the final outbreak size; $S(\infty)$ is the fraction that escaped infection.

The basic reproduction number is: $R_0 = \beta/\gamma$.

It is the expected number of secondary infections caused by one infected case in a fully susceptible well-mixed population. If $R_0 > 1$, early growth is possible; if $R_0 < 1$, infections tend to die out.

Speaker notes

This slide now comes after the aggregate definition. The well-mixed model assumes everyone can contact everyone in the same way. On an actual network, contacts are edges. The curve gives three important quantities: outbreak threshold, epidemic peak, and final size.

SIR: Closed Form or Approximation?

The standard well-mixed SIR equations usually do **not** give a simple closed-form expression for $S(t)$, $I(t)$, and $R(t)$ over time.

> In practice, we usually compute SIR curves by numerical integration of the ODEs, meaning ordinary differential equations, or by stochastic simulation on a graph.

> This does not mean "theory is useless." It means analytical results usually answer specific questions rather than giving the full curve in one formula.

Speaker notes

This slide prevents students from expecting a neat formula for the full epidemic curve. Use it as a bridge: we cannot usually write down $S(t)$, $I(t)$, and $R(t)$ explicitly for all t, but we can still derive useful properties of the outbreak.

Analytical Results: Threshold and Peak

Threshold intuition

At the beginning, almost everyone is susceptible. The central question is:

> Does one infected case create more than one new infected case on average?

In the simplest well-mixed SIR model, this is summarized by $R_0 = \beta/\gamma$.

If $R_0 > 1$, early growth is possible. If $R_0 < 1$, the outbreak tends to die out.

Peak condition

The infected curve peaks when it stops increasing:

$$\frac{dI}{dt} = 0.$$

For well-mixed SIR:

$$\frac{dI}{dt} = \beta SI - \gamma I.$$

So the peak occurs when infection pressure and recovery balance:

$$\beta S = \gamma.$$

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Speaker notes

This slide gives two analytical uses of the ODE without solving the full trajectory. The threshold is about early growth. The peak condition is about when current infections stop rising. These are actionable even when the complete curve must be computed numerically.

Analytical Results: Final Size and Uncertainty

Final-size relation

Even without a closed form for the whole curve, SIR can characterize the final outbreak size implicitly. Implicit means the result is defined by an equation that usually still needs numerical solving.

Definition

Final outbreak size is defined as:

$$R(\infty) = \frac{|i \in V : X_i(\infty) = R|}{|V|} = \frac{|i \in V : i \text{ was ever infected}|}{|V|}.$$

In words: count the nodes that end in state R and divide by the number of nodes.

Here **removed** means "out of the spreading process": no longer infectious and not susceptible again in this SIR model.

Usually $R(\infty) < 1$ because some susceptible nodes are never reached:

$$S(\infty) + R(\infty) = 1.$$

So final size is not current infection. It is cumulative reach.

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Speaker notes

This slide completes the answer to "what do analytical results give us?" The likely confusion is that "final infected" is zero in SIR, because the I compartment empties. Final size means cumulative recovered/removed, R(infinity), not I(infinity). It can be less than one because transmission dies out before reaching all susceptible nodes, especially below threshold, in modular graphs, or by stochastic extinction. Final size is often available only implicitly, while stochastic simulation gives distributions rather than a single curve.

Simulation or Closed Form?

There are three common computational levels:

Level	Representation	What it gives
stochastic network simulation	state of every node	sample paths, uncertainty, seed effects
aggregate ODE (ordinary differential equation)	counts/fractions $S(t), I(t), R(t)$	smooth curves, thresholds, peak intuition
network approximation	degree moments, eigenvalues, percolation	topology-sensitive thresholds/final size

> Closed-form solutions are limited. For real networks, simulation is the default; formulas are used to understand regimes and check intuition.

Speaker notes

This slide directly addresses how the models are used. The right answer is not "simulate or solve" universally. It depends on the question. For teaching, make students map the research question to the computational level.

Network Simulation Template

For SI, SIS, and SIR, the simulation pattern is nearly the same:

```
initialize node states
for t = 1 ... T:
  collect infection attempts over S-I edges
  apply successful infections with probability beta
  apply recovery/removal with probability gamma
  record S(t), I(t), R(t)
```

> The model-specific part is the post-infection state: SI keeps nodes infected, SIS returns them to susceptible, SIR moves them to recovered/removed.

Speaker notes

This slide makes the computer science detail explicit. It also clarifies why the same implementation skeleton can support all three models. Mention that event-based simulations such as Gillespie-style algorithms are often used for continuous-time versions; synchronous updates are easier for teaching and classroom examples.

Simulation Stability: What To Expect

Stochastic epidemic simulations are not single deterministic answers. Repeated runs can differ, especially near the epidemic threshold.

Model	Expected long-run behavior
SI	monotone growth until no reachable susceptible nodes remain
SIS	extinction or noisy persistence around an endemic level
SIR	one outbreak wave, then absorption when $I(t) = 0$

> Near a threshold, small random differences can decide whether the infection dies early or becomes large. Report several runs, averages, and uncertainty bands rather than one trajectory.

Speaker notes

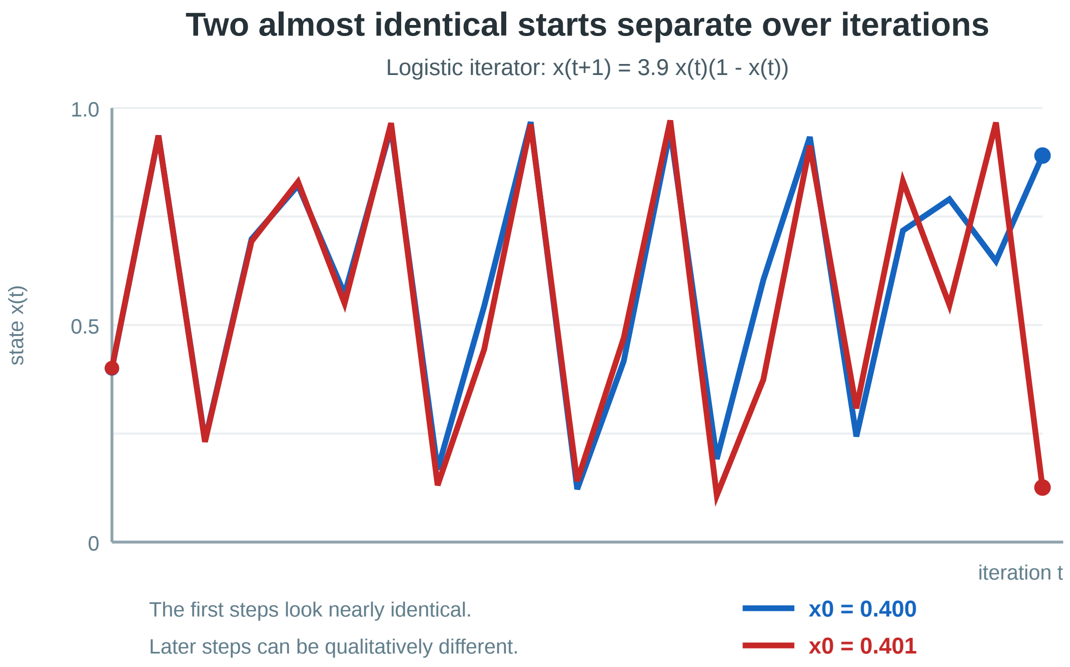
This slide explains simulation stability in student-friendly terms. SI and SIR have absorbing endpoints in finite graphs. SIS also has an absorbing all-susceptible state in finite stochastic simulations, but above threshold it may persist for a long time around a quasi-stationary endemic level before eventual extinction.

Numerical Stability: Iteration Can Amplify Error

Even a simple iterative rule can become numerically sensitive:

$$x_{t+1} = r x_t (1 - x_t).$$

For some values, such as $r = 3.9$, two almost identical starting values can separate quickly (May, 1976).



> This logistic iterator is not an epidemic model. It is a warning: simulations are repeated numerical updates, and repeated updates can amplify rounding, step-size, and modeling choices.

Speaker notes

Use this as a visual computational caution. The point is not to say SIR is chaotic, but to make students respect iterative simulation. If a result changes when the time step or implementation details change, the simulation result is not yet robust.

Reproducibility: Seeds, Precision, Hardware

> Reproducibility also depends on random seeds. Fellicious, Weissgerber & Granitzer (2020) show this for CNN accuracy: reporting a single run can hide relevant variation caused by initialization and training randomness.

> On GPUs and parallel hardware, precision and operation order add another layer: floating-point reductions can produce slightly different results even when the mathematical formula is the same (Goldberg, 1991; Higham, 2002).

For simulation studies, this means:

- the random seed controls one possible trajectory through the stochastic process;
- the time step controls how the continuous process is discretized;
- CPU/GPU and precision settings can change accumulated rounding behavior;
- parallel update order can change results when operations are not exactly associative.

> A robust claim should survive reasonable changes in seed, step size, precision, and hardware.

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Speaker notes

The Fellicious, Weissgerber and Granitzer paper is a good local example for random seed effects in neural network experiments. GPU execution adds further concerns through precision and parallel operation ordering. Keep the message practical: students should document seeds, run several repetitions, and test whether conclusions survive reasonable implementation changes.

Numerical Stability: Practical Checks

For epidemic simulations, check robustness before interpreting a single curve:

Check	Why it matters
repeat many runs	separates typical behavior from one random path
report uncertainty bands	shows threshold sensitivity and stochastic spread
vary time step or update rule	catches discretization artifacts
keep probabilities in $[0, 1]$	prevents invalid numerical states
compare CPU/GPU or precision settings	detects hardware and floating-point effects
record random seeds	makes results reproducible
compare to aggregate intuition	catches obvious implementation errors

> Numerical stability is not just "no crash." It means the qualitative conclusion survives reasonable changes in resolution, seed, and implementation choices.

Speaker notes

This is the practical computer science slide. It connects stochastic simulation to reproducibility and validation. The checks are intentionally simple enough for students to apply in exercises.

Module Takeaway

> Epidemic models are diffusion models with explicit state transitions. The crucial modeling decision is state memory: infected forever (SI), recover and become susceptible again (SIS), or recover into removal/immunity (SIR).

Effects of Network Topology

> **Guiding Question:** How does the same spreading rule behave on different graph families?

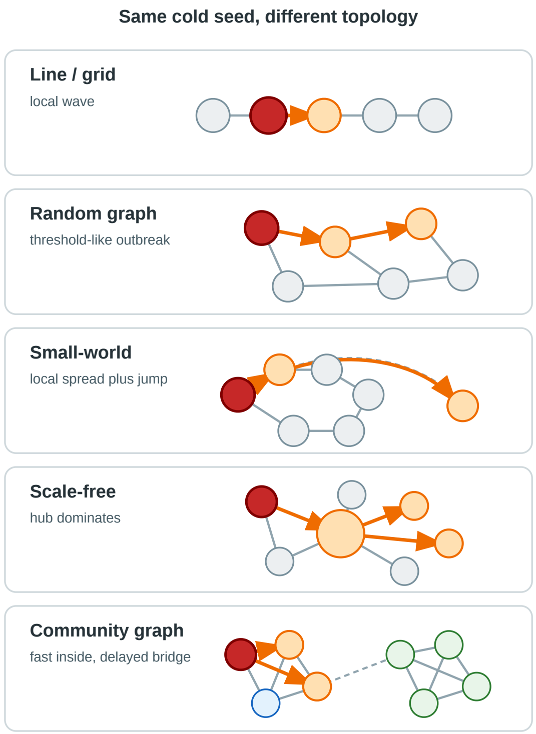
Speaker notes

This module is where the lecture connects back to L04 and L07 most strongly. We are not introducing topology from scratch. We are asking what previously learned structures do to diffusion.

Intuitive Example: A Cold Starts at One Node

Imagine the same cold with the same β and γ starts in five different networks:

Topology	First intuition
line or grid	slow wave
random graph	threshold-like outbreak
small-world	local spread plus long jumps
scale-free	hubs dominate
community graph	fast inside, delayed between groups



Speaker notes

This slide should be verbal and visual in the classroom. Students already know line, random graph, small-world, scale-free, and community graph from earlier lectures. The point is to forecast the epidemic behavior before formulas.

Homogeneous Networks: SIR

In relatively homogeneous networks, most nodes have similar degree. Average degree is therefore a useful first approximation: unlike a well-mixed model, contacts are still edges, but $\langle k \rangle$ summarizes the typical number of neighbors that one infectious node can try to infect.

For **SIR-like outbreaks**, average degree multiplies infection opportunities:

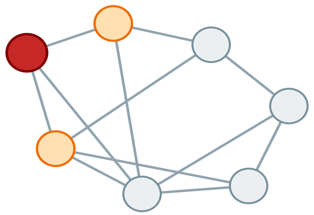
$$R_0 \approx \frac{\beta}{\gamma}\langle k \rangle.$$

Read this as an early-outbreak approximation:

- β controls how often infection succeeds along contacts.
- γ is the recovery rate; with continuous-time exponential recovery, the expected infectious time is $1/\gamma$.
- $\langle k \rangle$ is the typical number of contacts available.

If this product is larger than one, one infected node tends to create more than one new infected node before recovery.

Homogeneous graph: average degree is informative



No extreme hub: most nodes have comparable contact opportunities.

For SIR, the main use is outbreak risk and approximate final size: below the threshold most introductions die out; above it, a large recovered/removed set can become possible.

Speaker notes

Keep the formula as an intuition, not a universal law. The SIR expression says expected secondary infections scale with infectiousness times contact opportunities. In continuous time, gamma is a recovery rate and the exponential waiting time has mean 1/gamma. In a discrete-time simulation where gamma is a recovery probability per step, the expected duration is also about 1/gamma for a geometric waiting time, but the exact convention should be stated. The approximation assumes a relatively homogeneous graph and ignores clustering, degree correlations, and repeated contacts. Fully well-mixed models assume everyone can contact everyone in an aggregate rate; homogeneous-network models keep edges but summarize them by average degree.

Homogeneous Networks: SIS (1/2)

SIS has a different question than SIR. Because recovered nodes become susceptible again, we do not mainly ask "how large was the final outbreak?" We ask whether infection can persist as a long-running endemic process.

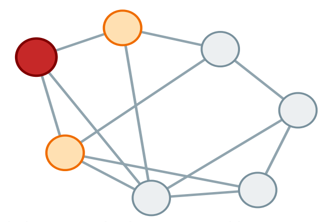
For a homogeneous graph, the aggregate approximation becomes:

$$\frac{\beta}{\gamma} \gtrsim \frac{1}{\langle k \rangle}.$$

This is the well-mixed SIS condition rearranged for a network with about $\langle k \rangle$ contact opportunities per node: persistence becomes plausible when infection strength times average degree exceeds recovery.

If $\langle k \rangle$ is larger, each infected node has more chances to reintroduce infection before recovery. Therefore the infection rate needed for persistence is lower.

Homogeneous graph: average degree is informative



No extreme hub: most nodes have comparable contact opportunities.

Homogeneous Networks: SIS (2/2)

For a general network, average degree can be too crude. A common network threshold intuition is:

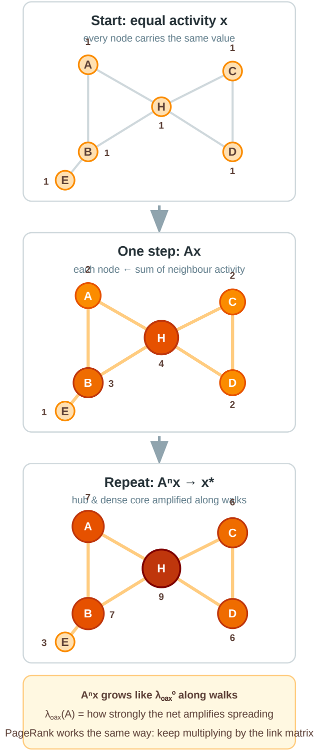
$$\frac{\beta}{\gamma} \gtrsim \frac{1}{\lambda_{\max}(A)}.$$

Here $\lambda_{\max}(A)$ is the largest eigenvalue of the adjacency matrix. For its leading eigenvector x:

$$Ax = \lambda_{\max} x.$$

Formally, $\lambda_{\max}$ is the largest factor by which multiplying a nonzero activity vector by A can stretch it.

Since Ax replaces each node's activity by the sum of activity coming from its neighbors, repeated multiplication models repeated spreading along walks. If the graph contains hubs, dense cores, or many mutually reinforcing paths, this neighbor-summing grows strongly. That is why $\lambda_{\max}(A)$ measures how much the network can amplify reinfection. This is exactly the power iteration you already know from **PageRank**, where the same repeated matrix multiplication converges to the leading eigenvector.



> An **endemic** SIS process means the infection does not simply burn out; it remains present for a long time at a nonzero infected fraction, even though individual nodes recover and become susceptible again.

Speaker notes

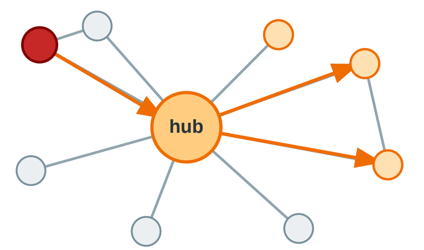
This slide merges the previous aggregate-network SIS slide with the homogeneous-topology intuition. The exact stochastic SIS process on a finite network can still die out by chance, but the threshold approximation explains whether persistence is structurally likely. Connect lambda_max to eigenvector centrality and PageRank intuition: repeated walks are amplified when the graph has dense cores or hubs.

Heterogeneous Networks and Hubs

In heterogeneous networks, the average degree can be misleading because infections are more likely to reach high-degree nodes.

> For SIR on uncorrelated random networks, a common threshold approximation uses **transmissibility** T: the probability that infection crosses an edge before recovery.
>
> $$T_c \approx \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}.$$
>
> For a graph with n nodes and degrees k_i:
>
> $$\langle k \rangle = \frac{1}{n}\sum_{i=1}^{n} k_i, \qquad \langle k^2 \rangle = \frac{1}{n}\sum_{i=1}^{n} k_i^2.$$
>
> T_c is a **network-level** critical edge-transmission probability. If the disease transmissibility T is above this threshold, a large outbreak becomes possible.

Heterogeneous graph: hubs lower thresholds



A hub creates many chances for onward spread and reinfection.

> This explains why hubs matter for SIR final outbreak risk: high-degree nodes increase $\langle k^2 \rangle$, which lowers the critical transmissibility T_c. Lower T_c means that even weaker per-edge transmission can cross the outbreak threshold.

Speaker notes

This slide clarifies T_c. For SIR, T_c is not R0 itself and not a per-node score; it is a graph-level threshold for the per-edge transmissibility T under a simplifying random-network approximation. Connect it to the previous percolation slide: T is the probability that an edge is open as a successful transmission edge, and T_c is the point where a giant reachable component becomes possible. Individual nodes can still have very different local outbreak impact depending on degree and position.

Vanishing Threshold: What It Means

Pastor-Satorras & Vespignani (2001) showed that idealized infinite scale-free networks with sufficiently heavy-tailed degree distributions can have a **vanishing epidemic threshold**.

Mathematical idea

If the second moment $\langle k^2 \rangle$ grows without bound, then thresholds such as

$$T_c \approx \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}$$

can approach zero.

Interpretation

There is no positive critical value that the disease must exceed in the infinite-network limit.

Even weak transmission can be above threshold because hubs create many repeated spreading opportunities.

> This does **not** mean parameters such as β, γ, or R_0 no longer matter. They still affect speed, prevalence, and finite-network outbreak size. The result says the critical threshold can become extremely small in idealized hub-dominated networks.

Speaker notes

This slide prevents the common overstatement that scale-free networks spread epidemics independently of R0. A better phrasing is: in the infinite idealized limit, the threshold can tend to zero, so any positive transmission strength may be above threshold. Real networks are finite and constrained, so the threshold is usually small, not literally zero.

Reading the Hub Result Correctly

What it says

- Hubs are infected early with high probability.
- Hubs create many onward transmission opportunities.
- Targeted hub protection can be far more effective than random protection.

What it does not say

- Every real disease infects everyone.
- Every social network is scale-free.
- Control is impossible.

> The practical lesson is targeted intervention, not fatalism.

Speaker notes

This slide is important because scale-free epidemic claims are often overgeneralized. Make students distinguish model result, real network, and policy implication.

Theory: Clustering, Shortcuts, and Communities

Structure	Effect on simple contagion	Effect on threshold / complex contagion
local clustering	repeated contacts, local saturation	reinforcement, easier adoption
weak tie / bridge	fast jump to new group	often too thin to trigger adoption
small-world shortcut	collapses path length	helps only if reinforcement follows
community boundary	bottleneck for global spread	may trap adoption inside modules

> Watts & Strogatz (1998) explain how few shortcuts can sharply reduce distances while preserving clustering. Centola (2010) shows that for complex contagion, clustering can beat shorter paths.

- **Simple contagion:** one successful exposure can be enough to transmit, as in a cold or SI/SIR infection.
- **Threshold / complex contagion:** one exposure is usually not enough; adoption needs reinforcement from several neighbors, as in the linear threshold model.

Speaker notes

This table is the core bridge between L07 and L08. Simple contagion means a single exposure can trigger transmission. Threshold or complex contagion means social reinforcement matters, so one bridge may not be sufficient. Small-world shortcuts are excellent for simple contagion because one contact can seed a new region. For threshold behavior, shortcuts alone may not be enough; local reinforcement matters.

Practical Example: Intervention by Topology

The same infection model suggests different interventions once the contact graph changes.

Topology	What to look for	Intervention logic
grid / spatial contact	local infection front	contain neighborhoods around the boundary before the wave expands
random graph	many similar contacts	reduce average transmissibility T or average degree $\langle k \rangle$
scale-free / hub-heavy graph	high-degree nodes and venues	protect hubs first because they lower the epidemic threshold
community graph	sparse bridges between groups	monitor or reduce bridge contacts before infection jumps modules

> The intervention should match the structural bottleneck. A grid is slowed by spatial containment; a hub-heavy graph is slowed by protecting hubs; a modular graph is slowed by controlling bridges.

Speaker notes

This table translates theory into action. It also reinforces that "remove central nodes" is not a universal answer. On a grid, there may be no special hub to remove. On a random graph, broad reduction in transmissibility can matter more than targeting one node. On a modular graph, a low-degree bridge can be more important than a high-degree node inside a community.

Applications: Topology Changes the Question

Topology changes what is worth asking. Before fitting a model, identify which structural feature is likely to dominate the spread.

Applied setting	Structural feature to inspect	Question topology adds
public health	households, workplaces, hubs, communities	Will the outbreak stay local, jump communities, or become system-wide?
online misinformation	bridges between groups and high-reach accounts	Which accounts or bridges turn local exposure into a cascade?
organizational communication	bottlenecks and redundant channels	Does information depend on one bridge, or can it reroute through alternatives?
product adoption	clustered communities and wide bridges	Is one exposure enough, or does adoption require local reinforcement?
infrastructure failure	dependencies and cascade paths	Which failures disconnect or overload large parts of the system?

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> The modeling question becomes: **what spreads, what state memory does it have, and which structural feature controls reach, speed, or persistence?**

Speaker notes

The applications should feel like modeling prompts. Ask students: What are the nodes? What are the edges? What spreads? Is one exposure enough? Does the state remain, recover, or become removed? Which topology is likely to dominate the result? The point is not that every application has one topology, but that the dominant topology determines the useful question.

Module Takeaway

Topology does not merely change the speed of spread. It can change the phase behavior: whether a process dies out, explodes, persists, stalls at a community boundary, or remains local because long-range paths are missing.

Examples and Applications

> **Guiding Question:** How do we choose a diffusion model for a real case?

Speaker notes

This final module consolidates the lecture. It keeps the empirical anchor from the old L08 but places it after students have the conceptual machinery.

Intuitive Example: Model Choice Is a Claim

When we choose a diffusion model, we are not just choosing a formula.

> We are claiming what kind of exposure changes a node.

If you believe...	Use...
one exposure can trigger spread	simple contagion / independent cascade
repeated social proof is necessary	threshold / complex contagion
infection can return	SIS
infection produces immunity/removal	SIR
attention flows continuously	random-walk or damped diffusion

Speaker notes

This slide is a useful checkpoint near the end. It also trains students to critique modeling papers: what behavioral assumption is hidden in the diffusion rule?

A Compact Model Map

Model	Node state	Transition rule	Typical output
Damped diffusion	continuous score	matrix update with restart	ranking / relevance
Independent cascade	inactive / active	one probabilistic chance per edge	final adoption set
Linear threshold	inactive / active	weighted exposure exceeds threshold	cascade size
SI	susceptible / infected	infection without recovery	cumulative reach
SIS	susceptible / infected	recovery allows reinfection	persistence
SIR	susceptible / infected / removed	infection then removal	peak and final size

Speaker notes

This table is the lecture's main synthesis artifact. It is also useful for exam preparation and exercise work.

Empirical Anchor: Centola's Experiment

> **Centola (2010),** *The Spread of Behavior in an Online Social Network Experiment*, Science 329, 1194-1197.
>
> Centola built an **artificial online health community** and randomly assigned 1,528 participants to anonymous "health buddies." The spreading "behavior" was **voluntarily signing up to use a new online health resource** — an adoption that takes effort, so people tend to join only after **several** neighbors have, not after just one.

The design

- Each participant had a **fixed set of neighbors** of equal size in both conditions.
- When a neighbor adopted, the participant got an **email signal**. More adopting neighbors → more signals.
- Only the **wiring** differed, so any difference in spread is caused by structure, not by degree, content, or incentives.

Two matched topologies

Condition	Structure	Prediction
clustered lattice	neighbors share neighbors; high clustering	more local reinforcement → faster spread
random	same degree, ties rewired; short paths, low clustering	shortcuts but little reinforcement → weaker spread

Speaker notes

The two networks are matched on everything except clustering: same number of nodes, same degree for every node, identical recruitment, content, and anonymity. The random network is the clustered lattice with its ties rewired (a Watts-Strogatz-style move), so it has shorter average paths but neighbors rarely overlap. The reinforcement mechanism is the email signal: a participant whose buddies adopt receives repeated, independent nudges, which is exactly the "two or more exposures" requirement of complex contagion. The critical design point is causal identification: because structure was randomly assigned, Centola could isolate clustering's effect on behavior spread rather than merely correlate them.

What Centola Found

Centola found that adoption spread farther and faster in clustered networks than in random networks.

> For behavior spreading, clustering beat shortcuts.

Why this matters:

- random networks had shorter paths, which should help simple contagion
- clustered networks supplied repeated social exposure
- adoption was **dose-dependent**: each additional adopting neighbor raised the chance a participant adopted — the signature of reinforcement
- the result supports the complex-contagion interpretation for this behavior

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Speaker notes

Centola's study is narrow but powerful. It does not prove that all behaviors are complex contagions. It shows that when adoption requires reinforcement, clustered networks can outperform random-shortcut networks despite longer paths.

What the Study Could Not See

Controlled networks are not natural networks. Real social networks are heterogeneous, layered, temporal, and adaptive.

The behavior was binary. Registration was observed, not intensity or long-term engagement.

The process had no reversal. Once registered, participants stayed registered in the measured outcome.

The result is about one behavior. Other behaviors may be simpler, more private, cheaper, or more reversible.

Speaker notes

This follows the empirical-anchor pattern in the earlier lectures. Clean causal designs usually narrow the setting. That is not a weakness; it is the tradeoff that makes the result interpretable.

Applications: Pick the Model Before the Metric

Situation	First modeling question
vaccination strategy	SIR or SIS? who are hubs and bridges?
misinformation	one exposure or repeated exposure? directed graph?
health campaign	threshold? which groups need reinforcement?
Web ranking	which random-walk distribution is meaningful?
supply-chain failure	directed dependencies? irreversible failures?

Speaker notes

This is the final practical translation. The message is: do not start with centrality. Start with the process. Then decide which structural measures matter.

Module Takeaway

> Diffusion modeling is model selection. Before computing a metric, decide what exposure means, whether the state sticks, and which topology the process can actually use.

Wrap-Up

Question	Modeling choice
What spreads?	signal, attention, behavior, infection, failure
What changes a node?	flow, one exposure, threshold, infection
Does it stick?	SI, SIS, SIR, behavior change, decay
Which topology matters?	hubs, bridges, clusters, communities, direction, timing
What is the outcome?	score, reach, peak, persistence, final size

The same graph can accelerate, block, localize, or sustain a process. Network dynamics require matching the spreading rule to the structure.

Speaker notes

This is the final synthesis. It should leave students with a method rather than a list of models: define the process, choose the state rule, inspect the topology, then interpret outcomes.

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Speaker notes

These references ground the non-basic claims in the lecture. The slide uses the same author-year style as the rest of the course and includes DOI links where they are central sources.